Ontologias, Gêmeo Digital e Telemetria Inteligente para a Infraestrutura da Internet (DT-II)

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Abstract

This project proposes a semantic unification of the Internet infrastructure through the employment of a Top Ontology, grounded on the Unified Foundational Ontology (UFO) and formalised in the axioms of Description Logic (TBox, ABox and RBox), using the W3C Web Ontology Language (OWL). The approach relies on modelling tools such as OntoUML, which allow the use of mereology and relators to constitute a "Logical Constitution" capable of transcending specific technological implementations (such as wireless networks, fibre optics or satellite systems). The central elements are organised into a fundamental triad composed of Network Node, Connection and Channel. The integration of this rigorous model into a dynamic Knowledge Graph, associated with inference mechanisms, ensures universal semantic interoperability and enables automated fault detection, cascading-impact analysis and advanced governance management, including Service Level Agreements (SLAs) and Semantic Jurisdiction. The outcome is a Digital Twin model that transforms passive infrastructure into an autonomous and resilient system, able to provide operational intelligence and support real-time decision-making.

Keywords domain ontology · upper ontology · ufo · ontouml · netconf · digital twin

1 Introduction

The relationship between ***Ontology*** and the ***Knowledge Graph*** (KG) is central to understanding how modern ***Artificial Intelligence*** (AI) organises information Kejriwal et al. [2021], Landgrebe and Smith [2025], Braga et al. [2025]. In the context of Internet Infrastructure (II), the ontology provides the logical framework, while the KG represents its dynamic instantiation. This project proposes a ***Upper Ontology*** (UO) for II Arp et al. [2015], structured on four pillars:

P1. **Theoretical Foundation**: The ontology defines classes, relations and axioms, while the KG organises facts in triples `Subject-Predicate-Object`. ***Description Logic*** (DL) Baader et al. [2017], Nardi and Brachman [2010], Horrocks et al. [2006] ensures computational rigour for hierarchies and constraints.
P2. **Modelling Methodology**: Using the ***OntoUML Lightweight Editor*** (OLED) Guerson et al. [2015], Guizzardi et al. [2018], the ontology distinguishes essence (*Kind*) from role (*Role*), exporting models in interoperable formats such as ***Turtle*** and ***OWL*** W3C OWL Working Group [2012].
P3. **Upper Ontology**: A **Logical Constitution** unifies subdomains (for example, 5G, satellites, optical fibres) in the triad: **Node**, **Connection** and **Channel**.
P4. **Operational Intelligence**: A ***Digital Twin*** Grieves and Vickers [2017], Zhou et al. [2026], operated by an ***Inference Engine***, integrates ***TBox***, ***ABox*** and ***RBox***, fundamental components of Description Logic, to detect faults and manage governance, including ***Service Level Agreements*** (SLA) and semantic jurisdictions[1].

1.1 A practical roadmap for the DT-II project

DT-II is the name assigned to this project. This section describes in detail its various components. Figure 1 summarises the project's **Intelligence Cascade**.

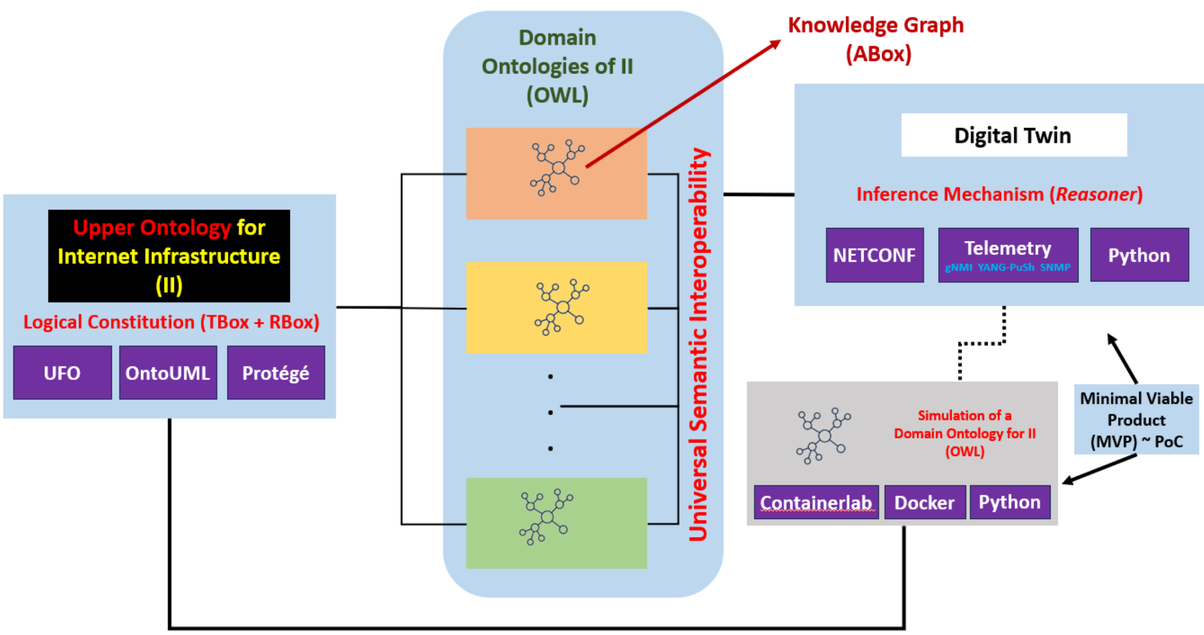

Figure 1: Overview of the DT-II Project Architecture

[1]Semantic Jurisdiction is understood as a normative domain of logical authority, formally established by ontological axioms, which regulates the behaviour of digital entities according to their meanings, contexts and relations. Unlike geographical boundaries or physical attributes, it is a governance layer mediated by relators, capable of propagating constraints and norms (such as legislation, service-level agreements and privacy policies) among nodes, connections and channels of an infrastructure. Its function is to ensure inter-domain compliance through automated reasoning, allowing rules to be checked, inherited and applied dynamically in data flows and network operations, guaranteeing normative integrity in complex and changing environments.

The figure shows how the UO, the DOs and the Digital Twin interact to ensure semantic interoperability[2] Guizzardi and Guarino [2024] and promote dynamic governance, also characterising the test environment and the *Minimal Viable Product* (MVP), in this case representing the Proof of Concept (PoC).

On 04/06/2026, the Internet had approximately **78,809** active Autonomous Systems (ASes), according to the **IPv6 CIDR Report**[3] maintained by Geoff Huston. Each AS has operational autonomy, with its functionality established by protocols and specialised administrators. The correctness of this functionality characterises the behaviour of the Internet from the user's perspective. A misconfiguration in an AS can propagate through the global Internet infrastructure, causing incidents with worldwide reach. For example, if an AS wrongly announces an IPv4 block via the BGP protocol, it can cause serious harm, particularly to the legitimate holder of that block. One way to prevent anomalous behaviours in the AS environment is through measurements known as **telemetries**.

These risks highlight the need for advanced monitoring mechanisms. The DT-II project proposes the simulation of ontology-based techniques (or knowledge bases) capable of establishing intelligent telemetries without human intervention. Associated with ontologies, protocols such as NETCONF and related technologies are used, widely deployed and implemented by major Internet infrastructure equipment vendors.

The project employs two types of ontologies: domain ontologies (DOs), which specify the knowledge of each AS, and the UO, which ensures semantic interoperability and facilitates the construction and verification of DOs through the OntoUML tool. Although DOs are similar across most ASes, the project's innovation lies in the use of the UO, which brings benefits such as:

- Guaranteeing semantic interoperability between different domains;
- Facilitating the systematic construction and verification of DOs via OntoUML.

Additionally, the project uses *Protégé*, supported by gUFO, for modelling the DOs both in the testbed and in the ASes participating in the second phase of the research. The testbed, developed with ***Containerlab***, ***Docker*** and ***Python***, constitutes the DT-II Proof of Concept (PoC), allowing validation of the project implementation.

The DT is the operational component of the project. It adopts the autonomic characteristics described in Murch [2004] and must behave by maintaining the eight elements proposed to preserve the characteristics of an autonomic computing system:

- Comprehensive knowledge of itself, encompassing available resources and operational capabilities
- The capacity to configure and reconfigure autonomously in response to contextual requirements
- The capability to continuously self-optimise, ensuring sustained efficiency and performance
- Robust self-healing mechanisms to detect, isolate, and recover from faults
- Self-protection capabilities to safeguard against internal inconsistencies and external threats
- The ability to acquire and interpret knowledge of its environment and context, adapting behaviour accordingly
- Competence to operate effectively within heterogeneous and distributed computing environments
- The ability to anticipate, interpret, and adapt to evolving user requirements

A DT can maintain an environment that preserves the above items — namely self-knowledge, self-configuration, self-optimisation, self-healing, self-protection, etc. — if conceived as a **multi-agent** architecture supported by a semantic knowledge base. To this end, the DT follows the **MAPE-K** proposal Kephart and Chess [2003], Corporation [2005], which establishes the following cycle as a means of ensuring autonomy:

- Monitoring (M): collects real-time data.
- Analysis (A): interprets the data based on axioms and rules.
- Planning (P): generates action plans consistent with the Logical Constitution.
- Execution (E): applies changes to the infrastructure.
- Knowledge (K): feeds back into the knowledge base, ensuring continuous learning.

The Figure 2 characterises with visual details the MAPE-K flow of the DT.

[2] Semantic interoperability refers to the ability to connect different information systems by accurately understanding how the real-world entities represented in those diverse systems relate to each other, going beyond simple packet exchange (technical interoperability) or data-format understanding (syntactic interoperability).

[3] https://www.cidr-report.org/

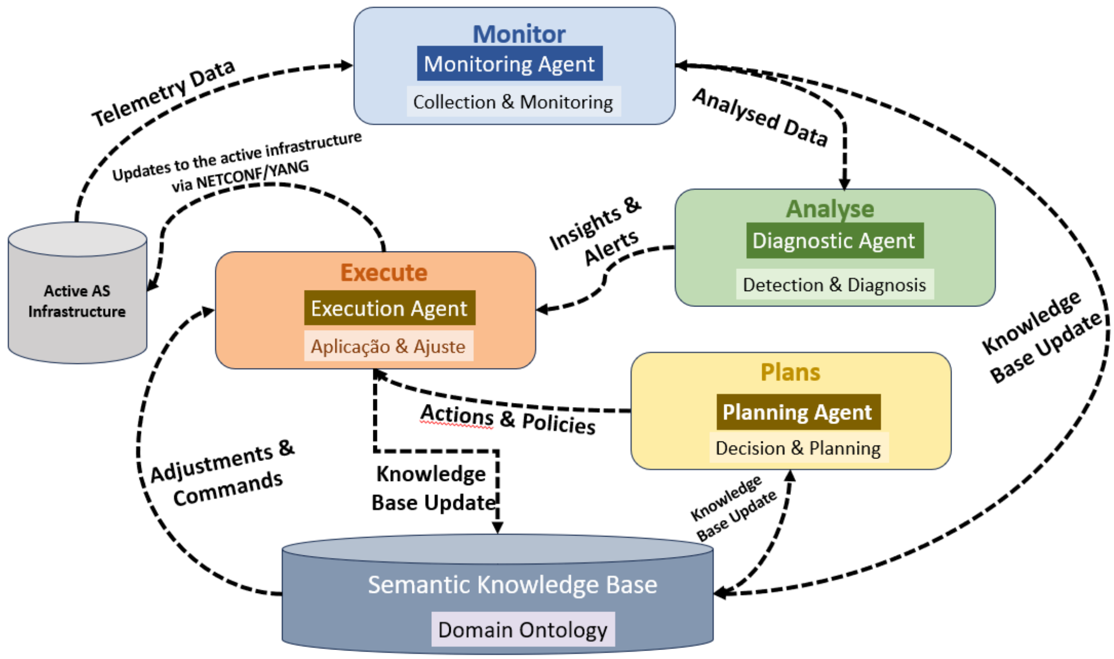

Figure 2: MAPE-K Flow of the Digital Twin

2 Methodology

This project follows the ontology engineering life cycle Keet [2025], Parks [2025], Effingham [2013], using the ***Unified Foundational Ontology*** (UFO) as a basis and OntoUML as the modelling language Guizzardi [2005]. The process is structured in four technical phases:

M1. **Conceptual Abstraction**: UFO theory is applied to classify network assets. Rigid essence classes (`PhysicalElement`, `MaterialMedium`) define identity, while accidental roles (`NetworkNode`, `CommunicationChannel`) are linked by Relators.

M2. **Formalisation in Description Logic (DL)**: Concepts are axiomatized in DL, ensuring computational interpretability. For example, every `NetworkNode` must have a logical identifier:

$$\text{NetworkNode} \sqsubseteq \text{PhysicalElement} \sqcap \exists \text{possesses.LogicalIdentifier} \tag{1}$$

M3. **Semantic Serialisation**: Models validated in ***OLED*** are exported to OWL and serialised in ***Turtle***. Standardised vocabularies such as ***Dublin Core*** (DC) and ***FOAF*** provide support for metadata and provenance.

M4. **Automated Reasoning**: Inference engines confront the ***TBox*** and the ***ABox*** to check consistency and detect anomalies. Predicate logic allows deduction of failure states; for example, an inactive Channel implies compromise of its logical connection.

2.1 Ontological Analysis Structure and Mapping Process

The mapping between the II domain and UFO, via its gUFO implementation, follows a specialisation protocol grounded on logical metaproperties. For each local domain entity, the **Ontological Analysis Structure** is applied Almeida et al. [2026, 2019], Guizzardi [2005]:

A. **Specialisation of Fundamental Types**: Entities with rigid identity and specific individuation principles (for example, Physical Routers, Satellites, Fibre Cables) are mapped as subclasses of *gufo:Kind*. These represent substantial objects that persist independently of their configuration state.

B. **Relational Specialisation (Roles)**: Concepts whose existence depends on a specific topological or functional context (for example, Edge Node, Peering Point, Transit AS) are characterised as *gufo:Role*. This allows the Digital Twin to capture the dynamic nature of network functions independently of the underlying hardware.

C. **Relational Foundations**: Logical connections and service agreements that establish dependencies among multiple objects (for example, BGP Sessions, SLA Agreements) are formalised as *gufo:Relator*. This ensures that the integrity of the Knowledge Graph is maintained through the existential dependency constraints inherent to gUFO, where the invalidation of a relator triggers the logical removal of associated roles.

By employing this structure, DT-II transcends mere data modelling, establishing a **Logical Constitution** in which network behaviour is governed by fundamental ontological axioms rather than purely heuristic or statistical patterns.

2.2 Dynamic Synchronisation via YANG-Push

A critical requirement for DT-II is high-fidelity synchronisation between physical infrastructure and its semantic representation. To achieve this, the model employs *Model-Driven Telemetry* (MDT) via the **YANG-Push** mechanism (RFC 8641 Clemm and Voit [2019]).

Unlike traditional query-based (*pull*) monitoring, YANG-Push allows Network Nodes to directly stream state changes to the *Semantic Ingestor*. This integration is mapped into the gUFO framework as follows:

- **Periodic Subscriptions**: Quantitative metrics, such as interface counters, are streamed to update `gufo:Quality` assertions in the ABox.
- **Change Notifications**: Critical events, such as oscillations in BGP sessions, are treated as `gufo:Events` that immediately trigger re-evaluation of existential dependencies in `gufo:Relators`.

This ensures that the Digital Twin's Logical Constitution always reflects the current operational state of ASes with sub-second latency.

3 Relations between Ontology, Knowledge Graph and Description Logic

The ***Knowledge Graph*** (KG) operationalises the ontology by linking **real data** (instances) to the conceptual model. It represents information as **nodes** (entities) connected by **edges** (relations). For example, the node `The Godfather` is linked to `Francis Ford Coppola` via `directed_by`. Typically, a KG integrates multiple sources to form a network of interconnected facts.

Ontology and KG are **complementary**: the ontology provides semantics and logical rules, while the KG instantiates them as a dynamic, queryable knowledge base. Table 1 summarises their roles in the proposed framework.

Table 1: Complementarity between Ontology and Knowledge Graph

Characteristic	Ontology	Knowledge Graph
Focus	Definitions, rules, logic.	Data, facts, connections.
Level	Abstract (the "DNA").	Concrete (the "Organism").
Function	Provides meaning and semantics.	Enables queries and discovery.

Description Logic (DL) underpins modern ontologies (for example, OWL on the Semantic Web) Baader et al. [2017]. DL enables the formal definition of concepts and computational reasoning. For example, the axiom:

$$\text{Director} \sqsubseteq \text{Person}$$

establishes that every Director is a Person. More complex definitions use logical connectives, such as:

$$\text{Filmmaker} \equiv \text{Person} \sqcap \exists \text{directed.Film}$$

meaning that a Filmmaker is a Person who has directed at least one Film. Here, $\sqcap$ denotes intersection (AND), ensuring precise semantic constraints.

In summary, the ontology provides the logical and semantic structure, while the KG materialises that structure in concrete data. DL acts as the formal link between them, ensuring consistency, interpretability and inferencing capability. This triad — Ontology, KG and DL — constitutes the methodological core of DT-II, enabling the Internet infrastructure to be represented, monitored and governed semantically and dynamically.

4 Conceptual Structure of the Digital Twin

This section presents a **reference architecture** for implementing the Internet Infrastructure Digital Twin (II) as an **operational KG** (***ABox***) governed by a ***UO*** (***TBox***, ***RBox***). The objective is to make the model executable: collect, normalise, validate, infer and support decisions. The implementation objectives are:

(a) **Unified semantic inventory**: represent nodes, connections and channels regardless of technology;

(b) **Continuous synchronisation**: align the KG with telemetry and topology changes;

(c) **Detection and diagnosis**: identify inconsistencies, faults and cascading impacts; and (d) **Verifiable governance**: encode and audit rules (SLA, jurisdiction, authorisation).

4.1 Reference Architecture

The architecture is organised into seven layers:

(i) **Ontological Layer (*TBox*)**: defines central classes (`NetworkNode`, `Connection`, `Channel`) and constraints;

(ii) **Identity Layer**: assigns stable IRIs to instances, separating identity from state;

(iii) **Ingestion Layer**: collects telemetry (YANG-Push, gNMI, SNMP, syslog) and normalises data into the Node–Connection–Channel triad;

(iv) **Persistence Layer (*ABox*, *RBox*)**: stores facts in RDF/OWL triplestores and keeps temporal history via observations;

(v) **Validation Layer**: applies structural constraints and logical consistency checks with DL reasoners;

(vi) textbfReasoning Layer: infers states (`DegradedConnection`, `SLAViolation`) and propagates failures;

(vii) **Exposure Layer**: provides SPARQL queries, dashboards, alerts and integration with NOC/orchestration systems.

4.2 Minimum Data Model

The initial implementation focuses on three entities:

(a) **Node**: identity (`hasIdentifier`), operational state, timestamp;

(b) **Connection**: endpoints (`connectNode`), supported channel, metrics (latency, loss, jitter);

(c) **Channel**: medium type (fibre, copper, radio), capacity, QoS policy.

4.3 Integration of *ABox* and *RBox*

The ***ABox*** instantiates infrastructure elements (routers, links, channels) from inventory data Claise et al. [2019]. The ***RBox***, based on $\mathcal{SROIQ}$ logic Horrocks et al. [2006], governs relational properties:

(a) **Transitivity**: deduces cascading failures;

(b) **Mereology**: validates part–whole relations for SLA compliance [Effingham, 2013, Chap. 8];

(c) **Property chains**: enable bidirectional navigation and anomaly tracing.

Together, ***ABox*** and ***RBox*** turn the KG into a computable model, enabling dynamic reasoning and governance.

4.4 Operational Flow

F1. Ingestion → normalisation → triple generation.

F2. Validation → integrity checks and anomaly detection.

F3. Inference → classification and rule-based derivations.

F4. Query/alert → recommendations or automated orchestration.

4.5 Example Scenario

In a fibre link failure:

(a) telemetry marks the `Channel` as unavailable;
(b) inference classifies the `Connection` as `InactiveConnection` and derives `SLAViolation`;
(c) redundancy is triggered (for example, satellite backup).

In summary, the Digital Twin evolves into an **executable knowledge system**, where the ***TBox*** governs meaning and rules, and ***ABox*/*RBox*** capture state, validate consistency and enable inference, turning data into actionable decisions.

5 Use Cases

5.1 Use Case I: Semantic Jurisdiction and Logical Governance in DT-II

Semantic Jurisdiction is the governance mechanism of the **DT-II** that establishes authority and policy compliance over digital entities based on meaning, context, and relations—transcending traditional geophysical boundaries. In this model, jurisdiction is no longer a static hardware attribute but a **normative entity mediated by Relators**.

5.1.1 Foundation and Axiomatization

Defined in the ***TBox*** through restriction axioms, Semantic Jurisdiction leverages the Reasoner to determine which norms (e.g., LGPD, GDPR, SLAs) apply to real-time data flows. In the ***RBox***, property chains formalize propagation: if a `NetworkNode` is bound to a legal norm, all `Channels` and `Connections` mediated by Relators inherit the same restrictions.

5.1.2 Role of the Digital Twin

The Digital Twin acts as custodian of jurisdiction. By ingesting telemetry (gNMI, YANG-Push, SNMP), the system populates the ***ABox***. Through **Semantic Elevation**, it verifies whether, for example, a health data packet traverses a node under a jurisdiction that fails to meet privacy requirements defined in the ***TBox***.

5.1.3 Operationalization and Resilience

In dynamic events (e.g., DDoS attacks, BGP failures), Semantic Jurisdiction guides the **MAPE-K** cycle Murch [2004]: (a) **Analysis**: Reasoner detects violations of jurisdiction axioms; (b) **Planning**: System searches for compliant alternatives; (c) **Execution**: Configuration is applied via ***NETCONF*** only after logical validation.

In summary, Semantic Jurisdiction transforms infrastructure into a ***legally aware*** system, where compliance is mathematically verified by DL, ensuring governance integrity in complex and mutable networks.

5.2 Use Case II: Behavior of the Model under a DDoS Attack

In a Distributed Denial of Service (DDoS) attack, response time is critical. The Digital Twin acts as a Semantic Immune System, leveraging the triad of ***TBox***, ***RBox***, and ***ABox*** within the ***MAPE-K*** cycle as follows:

- **Monitoring** Telemetry (gNMI, YANG-Push, SNMP) updates the ***ABox*** in real time. For example, a flow spike from 1 Gbps to 100 Gbps is recorded as:

  `inst:Unusual_Flow a ii:AnomalousFlow`

- **Analysis** The Reasoner applies ***RBox*** property chains: (a) the anomalous flow saturates `Channel_Alpha`; (b) transitivity propagates risk to dependent services (e.g., DNS); (c) the origin is traced to a specific **Semantic Jurisdiction**.
- **Planning** Consulting the ***TBox***, the system applies rules: if a critical service is at risk and the origin is external, mitigation (Blackholing or Rate Limiting) must occur at the edge. The plan prioritizes nodes sustaining the network's **Logical Constitution**.
- **Execution** Logical decisions are translated into technical commands. Via ***NETCONF***, edge routers are reconfigured to filter malicious traffic.

- **Advantages of the Semantic Approach** Unlike conventional defenses that block IPs, the **DT-II** enables: (a) **Context Awareness**: cooperation agreements mapped in the ontology may trigger governance APIs instead of simple blocking; (b) **Mereological Resilience**: the Reasoner may sacrifice non-vital channels to preserve critical services; (c) **Traceability**: each mitigation step is explainable, linked to axioms in the ***TBox***; (d) **Compliance**: no action is executed without validation by the ***Logical Constitution***.
- **Sequence Flow** The mitigation occurs within milliseconds, moving from the **Active Infrastructure** to the **Logical Core** and back. Figure 3 illustrates the process.

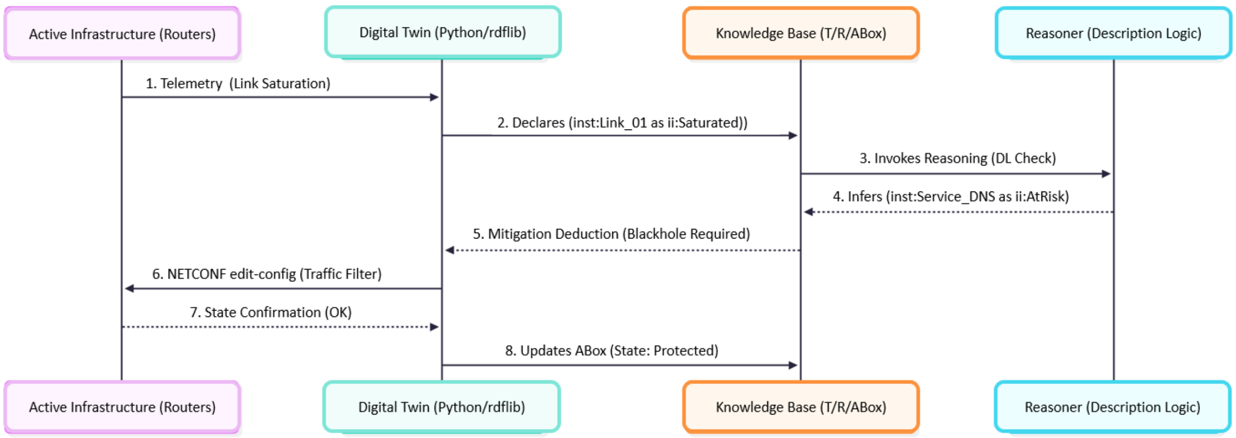

Figure 3: Sequence Diagram of DDoS Mitigation via DT-II

In summary, the ontology-based approach enables **policy-driven mitigation**, where assets of highest semantic and contractual value are prioritized, ensuring resilience and governance integrity.

6 Why Internet Infrastructure Requires a Project Like DT-II

The ***Internet Infrastructure*** (II) must be understood not merely as cables and protocols, but as a **living system** that has surpassed the limits of manual management. The **DT-II** project is necessary for four reasons: (R1) **From Static Management to Semantic Autonomy**: Current II management relies on rigid scripts and manual CLI commands. DT-II introduces **Logical Reasoning**, enabling the system to deduce solutions for complex failures and attacks under a ***Logical Constitution***; (R2) **NETCONF as Interoperability Catalyst**: The growing adoption of ***NETCONF*** provides a standardized, vendor-agnostic channel for the Digital Twin to interact with the **Active Infrastructure**, updating ***ABox*** instances in real time; (R3) **Global Semantic Governance**: In a fragmented regulatory landscape (e.g., LGPD, GDPR), DT-II codifies laws and SLAs as axioms, ensuring compliance regardless of geographic routing complexity; (R4) **Universal Semantic Interoperability**: Grounded in ***UO*** (***UFO***), DT-II resolves data silos, acting as a universal translator across domains (5G, Satellite, SD-WAN[4]), reducing costs and integration errors.

In summary, without DT-II, II remains an **Active Infrastructure**—processing packets without context. With DT-II, II gains **State Awareness**, evolving into an **Autonomic Network** that understands what it moves, why it moves, and whether it has logical permission to do so.

7 Conclusion and Future Works

This paper introduced the conceptual foundations of the **DT-II**, a Digital Twin for Internet Infrastructure grounded in Ontology, Knowledge Graphs, and Description Logic. By formalizing governance through **Semantic Jurisdiction** and enabling policy-driven responses to complex events such as DDoS attacks, the model advances the vision of transforming II from a passive infrastructure into an **autonomic, legally aware system**. The proposed architecture demonstrates how ontological rigor can ensure resilience, compliance, and semantic interoperability across heterogeneous domains. All results from this project will be made available at https://github.com/LJuliaoBraga/dtii/tree/main.

Future work will focus on:

[4]Software-Defined Wide Area Network, a software-based approach to WAN management.

(i) simulation of the Digital Twin to evaluate alarms in Autonomous Systems (AS);

(ii) experimental validation of spatiotemporal response mechanisms;

(iii) integration with programmable infrastructures via ***NETCONF***;

(iv) exploration of semantic interoperability across domains such as 5G, satellite, and SD-WAN;

(v) to participate in the evaluation of the two proposed telemetry alternatives—YANG-Push and gNMI—while also remaining open to those that may emerge in the near future; and

(vi) Integration with Quantum Key Distribution (QKD) and Quantum Internet: We plan to explore the use of Quantum Communication channels for the transmission of highly sensitive telemetry and governance metadata between the Active Infrastructure and the Digital Twin. As the DT-II handles critical "Semantic Jurisdictions" involving state-level privacy and strategic infrastructure topology, the integration of QKD will ensure that the synchronization between the physical and logical layers remains secure even against future quantum computing adversarial threats, maintaining the "Logical Constitution's" integrity through a provably secure quantum-classical hybrid architecture Kozlowski et al. [2023], Wang et al. [2024], Meter [2014].

These steps aim to evolve the design of the conceptual model to experimental deployment, culminating in its 2026 submission as a draft to the IRTF ***nmrg*** within the CGI-IETF initiative, in collaboration with the following Brazilian universities: **UFABC** (SP), **UFRGS** (RS), **IFSPe** (PE), and **UNICAP** (PE).

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Statement on Generative AI

During the preparation of this work, the authors used Gemini and Copilot exclusively for grammar and spelling checks in Portuguese and English. Support was obtained for translations and relevant evaluations. All substantive content, analysis and conclusions were produced by the authors, who reviewed and edited the text as necessary and assume full responsibility for the publication's content.

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